

Heng (Hugo) Zhao

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EDUCATION

Wake Forest University

B.S. in Computer Science and B.S. in Statistics | member of Upsilon Pi Epsilon | Cum Laude

John W. Sawyer Award: 2025, WFU CS Department, awardee

CRA Outstanding Undergraduate Research Awards: 2024-2025 Honorable Mention

Dean's List: 2021 Fall, 2022 Spring, 2022 Fall, 2023 Fall, 2024 Spring, 2024 Fall, 2024 Spring;

Wake Forest Research Fellowship: \$4,000 Recipient;

Undergraduate Honors Thesis: Episodic or Semantic Memory? An Empirical Study on Large Language Models (Computer Science), Structure and cohesion in word embeddings (Statistics)

Winston-Salem, NC

Aug 2021–May 2025

University of Virginia, School of Engineering & Applied Science

Ph.D. student in Computer Science

Charlottesville, VA

Aug 2025–Present

RESEARCH EXPERIENCE

Chen's Modeling and Learning Lab, Wake Forest University

Researcher advised by Dr. Minghan Chen

Winston-Salem, NC

Sep 2021–Oct 2022

• Conducted research on continual learning strategies for neural networks; adapted physics-informed neural networks (PINNs) for complex biological models.

Clinical Image Analysis Lab, WFU School of Medicine/Atrium Health

Center for Artificial Intelligence Research, internship advised by Dr. Mostafa Rezapour, and Dr. Bayram Annanurov

Winston-Salem, NC

Oct 2022–Dec 2023

• Explored machine learning applications in human organoid development and COVID-19.

Yang Research Group, Wake Forest University

Researcher advised by Dr. Fan Yang

Winston-Salem, NC

Sep 2023–May 2025

• Developed a Gaussian Concept Subspace (GCS) framework in collaboration with Dr. Fan Yang (WFU), Dr. Mengnan Du (NJIT), and Haiyan Zhao (NJIT). Co-authored 1 ICLR conference paper (accepted).

• Investigated parallels between human memory and LLM knowledge representation (CS honors thesis).

Berenhaut Research Group, Wake Forest University

Researcher advised by Dr. Kenneth S. Berenhaut

Winston-Salem, NC

Sep 2024–May 2025

• Studied aspects of structure, locality, density and distance in pre-trained word embeddings (Statistics honors thesis).

Tako Lab, University of Virginia

Graduate Researcher Assistant advised by Dr. Zhe Zeng

Charlottesville, VA

May 2025–Present

• Performs research in artificial intelligence and machine learning with focus on neurosymbolic AI and probabilistic ML

PUBLICATIONS

- H. Zhao, **H. Zhao**, B. Shen, A. Payani, F. Yang, M. Du, “Beyond Single Concept Vector: Modeling Concept Subspace in LLMs with gaussian Distribution,” ICLR, 2025
- **H. Zhao**, Z. Shao, G. Broeck, Z. Zeng, “SIMoE: A Probabilistic Framework for Cardinality-Constrained Routing in Mixture-of-Experts,” ICML, 2026

ORAL AND POSTER PRESENTATIONS

- “A Quick Peek to LLMs on Their Working Mechanisms,” URECA Day: A showcase of undergraduate scholarship and creative work, Wake Forest University, Winston-Salem, NC, September 27, 2024
- “CS Seminar I”, WFU CS department, Wake Forest University, Winston-Salem, NC, September 2, 2024

ORGANIZATIONS

Data Fest, Wake Forest University

Jan 2023– Mar 2025

Robotics Club, Wake Forest University

Aug 2021– May 2023

Chinese Students & Scholars Association, Wake Forest University

Aug 2021– May 2022

Asian Men's Basketball Team, Wake Forest University

Aug 2021– May 2025

ADDITIONAL SKILLS & INTERESTS:

Programming Languages: Python, C, Java, R, Unix Shell, LaTeX, MATLAB, STATA, SAS;

Interest: Jazz music; (alto/tenor/barton) saxophone; basketball, fencing, swimming